

MACHINE LEARNING ALGORITHMS

Practical Interview & Project Guide

Major algorithms • crisp concepts • when to use • why to use • examples • project practice

Designed for	Data Science / ML interviews and practical project building
Coverage	31 major algorithms and techniques
Learning style	Concept → decision rule → example → hands-on project

Important: “All machine-learning algorithms” is practically unbounded. This guide covers the major classical and practical algorithms you are most likely to encounter in real projects and interviews, plus core forecasting, recommendation, anomaly-detection, and neural-network methods.

1. How to Choose an Algorithm

Start with the **problem type**, then the data shape, then the business requirement. Do not choose an algorithm only because it is popular.

Question	Typical direction
Target is continuous?	Regression → Linear/Ridge/Lasso/Elastic Net, Trees, Random Forest, Gradient Boosting, XGBoost/LightGBM/CatBoost
Target is a class?	Classification → Logistic Regression, KNN, Naive Bayes, Trees, Random Forest, SVM, Boosting
No labels?	Clustering / dimensionality reduction / anomaly detection → K-Means, Hierarchical, DBSCAN, GMM, PCA, Isolation Forest
Text?	TF-IDF + Naive Bayes/Logistic Regression/SVM as classical baselines; transformers for modern NLP
Time series?	Naive baseline, Exponential Smoothing/Holt-Winters, ARIMA; then more advanced models as needed
Need recommendations?	Collaborative filtering / matrix factorization; combine with content and ranking methods
Need explainability?	Linear models and shallow trees are easier; use model-agnostic/feature-attribution methods for complex models

2. Universal ML Workflow

- **Define the target and business metric:** What exactly are you predicting or grouping?

- **Collect and inspect data:** types, missing values, duplicates, outliers, class imbalance, leakage.
- **Split correctly:** train/validation/test; for time series use chronological splits.
- **Build a baseline:** simple rule, mean/median, majority class, or simple model.
- **Preprocess:** impute, encode, scale when required, engineer features.
- **Train:** start simple, then increase model complexity.
- **Validate:** use the metric that matches the business cost.
- **Tune:** cross-validation or time-aware validation; avoid test-set tuning.
- **Explain:** feature importance/coefficient analysis/error analysis.
- **Deploy & monitor:** data drift, prediction drift, latency, performance, and retraining needs.

3. Common Metrics Cheat Sheet

Task	Core metrics	Use when
Regression	MAE, RMSE, R^2	MAE is easy to interpret; RMSE penalizes large errors more.
Balanced classification	Accuracy, F1, ROC-AUC	Accuracy is useful when class costs and frequencies are reasonably balanced.
Imbalanced classification	Precision, Recall, F1, PR-AUC	Use recall when missing positives is costly; precision when false alarms are costly.
Clustering	Silhouette, inertia + business profiling	No single metric proves clusters are useful; inspect stability and business meaning.
Forecasting	MAE, RMSE, sMAPE/MAPE	Use chronological validation and compare against a naive baseline.
Recommendation	Precision@K, Recall@K, MAP@K, NDCG@K	Evaluate ranked top-K recommendations, not only rating error.

A. Regression Algorithms

Linear Regression

Family	Supervised • Regression
Core idea	Fits a linear relationship between input features and a continuous target.
When to use	Numeric target with an approximately linear signal; useful as a baseline and for explainable relationships.
Why use it	Fast, interpretable, and easy to diagnose. Coefficients show the direction and magnitude of feature effects under the model assumptions.
Simple example	Predict house price from area, number of rooms, location score, and age.
Strengths	Simple; fast; interpretable; strong baseline.

Limitations	Sensitive to outliers and multicollinearity; misses nonlinear patterns unless features are engineered.
Practice project	House Price Prediction: clean a housing dataset, encode categorical variables, train Linear Regression, compare RMSE/MAE with tree models, and explain coefficients.
Useful metrics	MAE, MSE, RMSE, R^2 ; residual plots.
Interview cue	Know assumptions: linearity, independence, constant variance, low multicollinearity.

Polynomial Regression

Family	Supervised • Regression
Core idea	Extends linear regression with polynomial features such as x^2 and x^3 .
When to use	The relationship is curved but you still want a relatively simple parametric model.
Why use it	Captures smooth nonlinear relationships while retaining a familiar regression framework.
Simple example	Predict fuel consumption from engine size where the relationship bends rather than staying straight.
Strengths	Simple nonlinear extension; interpretable at modest degree.
Limitations	High-degree polynomials can overfit and become numerically unstable.
Practice project	Engine Efficiency Curve: compare degree 1, 2, 3, and 5 models using cross-validation and plot train/test error.
Useful metrics	RMSE, MAE, R^2 , cross-validation error.
Interview cue	Explain why feature scaling and regularization become important as degree grows.

Ridge Regression

Family	Supervised • Regression
Core idea	Linear regression plus L2 regularization that shrinks coefficients.
When to use	Many correlated features, or when ordinary linear regression overfits.
Why use it	Reduces variance and stabilizes coefficients while keeping all features.
Simple example	Predict salary using many correlated education, experience, skill, and role features.
Strengths	Stable; good with multicollinearity; useful baseline.
Limitations	Does not make coefficients exactly zero; requires tuning alpha.
Practice project	Salary Prediction with Correlated Features: compare Linear vs Ridge across alpha values using CV.
Useful metrics	RMSE/MAE/ R^2 and CV score.
Interview cue	Know the difference between L1 and L2 regularization.

Lasso Regression

Family	Supervised • Regression
Core idea	Linear regression plus L1 regularization; some coefficients can become exactly zero.

When to use	High-dimensional data where automatic feature selection is useful.
Why use it	Produces a sparse model and can remove weak predictors.
Simple example	Predict customer lifetime value from hundreds of behavioral features.
Strengths	Feature selection; compact model; interpretable.
Limitations	Can behave unstably with strongly correlated predictors; needs alpha tuning.
Practice project	Customer Value Feature Selection: train Lasso, inspect non-zero features, then retrain a compact model.
Useful metrics	RMSE/MAE/R ² ; number of non-zero features.
Interview cue	Explain why Lasso can select one feature among correlated features.

Elastic Net

Family	Supervised • Regression
Core idea	Combines L1 and L2 regularization.
When to use	High-dimensional data with correlated features where pure Lasso is unstable.
Why use it	Balances sparsity and coefficient stability.
Simple example	Predict insurance claim amount using many correlated customer and policy features.
Strengths	Feature selection plus stability under correlation.
Limitations	Two main hyperparameters; requires tuning.
Practice project	Insurance Claim Prediction: tune alpha and l1_ratio with cross-validation and compare Ridge/Lasso/Elastic Net.
Useful metrics	RMSE/MAE/R ² .
Interview cue	Know that l1_ratio controls the L1/L2 mixture.

B. Classification & Core Supervised Algorithms

Logistic Regression

Family	Supervised • Classification
Core idea	Models class probability using a logistic/sigmoid function.
When to use	Binary classification with a reasonably linear decision boundary and a need for probabilities/explainability.
Why use it	Fast, interpretable, strong baseline, and naturally produces probabilities.
Simple example	Predict whether a customer will churn: churn=1, stay=0.
Strengths	Fast; interpretable; probability outputs; strong baseline.
Limitations	Linear boundary unless features are transformed; sensitive to scaling and outliers.
Practice project	Customer Churn Classifier: preprocess categorical data, train Logistic Regression, tune threshold, and compare ROC-AUC vs F1.

Useful metrics	Accuracy, precision, recall, F1, ROC-AUC, PR-AUC, confusion matrix.
Interview cue	Know odds, log-odds, sigmoid, regularization, and threshold tuning.

K-Nearest Neighbors (KNN)

Family	Supervised • Classification/Regression
Core idea	Predicts from the labels/values of the nearest training points.
When to use	Small-to-medium datasets where local similarity is meaningful and a simple non-parametric method is acceptable.
Why use it	Very intuitive and can model nonlinear boundaries without explicit training.
Simple example	Classify a new fruit as apple/orange using nearest examples based on weight and color features.
Strengths	Simple; flexible; no explicit training phase.
Limitations	Slow prediction on large datasets; sensitive to scaling, noise, and choice of k.
Practice project	Product Similarity Classifier: scale features, test k=3..25, visualize decision boundaries, and evaluate CV performance.
Useful metrics	Classification: F1/accuracy; regression: MAE/RMSE.
Interview cue	Why scaling matters; how k affects bias/variance.

Naive Bayes

Family	Supervised • Classification
Core idea	Uses Bayes' theorem with a conditional-independence assumption between features.
When to use	Text classification, spam filtering, sentiment, and fast high-dimensional classification.
Why use it	Extremely fast and often surprisingly effective for sparse text features.
Simple example	Classify an email as spam or not spam from word counts.
Strengths	Fast; works well with high-dimensional sparse data; low data requirements.
Limitations	Independence assumption is often unrealistic; probability calibration can be imperfect.
Practice project	Spam Detector: TF-IDF + Multinomial Naive Bayes, compare with Logistic Regression, inspect most informative words.
Useful metrics	Precision, recall, F1, PR-AUC.
Interview cue	Know Gaussian vs Multinomial vs Bernoulli Naive Bayes.

Decision Tree

Family	Supervised • Classification/Regression
Core idea	Splits data using feature thresholds to form a tree of decisions.
When to use	Need a nonlinear, interpretable model with mixed feature types and little preprocessing.
Why use it	Captures interactions and nonlinear patterns and is easy to explain.

Simple example	Approve or reject a loan using income, credit score, age, and debt.
Strengths	Interpretable; nonlinear; little preprocessing; handles interactions.
Limitations	Can overfit badly; unstable to small data changes.
Practice project	Loan Approval: tune max_depth/min_samples_leaf, visualize the tree, and compare with Random Forest.
Useful metrics	Classification: F1/ROC-AUC; regression: RMSE/MAE.
Interview cue	Understand Gini impurity, entropy, information gain, and pruning.

Random Forest

Family	Supervised • Ensemble
Core idea	Averages many randomized decision trees for robust predictions.
When to use	Tabular data where you want a strong nonlinear baseline with less tuning than boosting.
Why use it	Reduces variance compared with a single tree and handles nonlinear interactions.
Simple example	Predict employee attrition from HR features.
Strengths	Robust; handles nonlinearities; useful feature importance; less overfit than one tree.
Limitations	Larger model; less interpretable than a single tree; can be memory-heavy.
Practice project	Employee Attrition: build a complete pipeline, tune n_estimators/max_depth/min_samples_leaf, inspect feature importance, and compare with XGBoost.
Useful metrics	F1, ROC-AUC, PR-AUC; RMSE for regression.
Interview cue	Bagging vs boosting; why random feature subsets reduce correlation between trees.

Extra Trees

Family	Supervised • Ensemble
Core idea	Builds highly randomized trees, including randomized split thresholds.
When to use	When you want a fast tree ensemble and extra variance reduction may help.
Why use it	Randomization can improve generalization and often trains quickly.
Simple example	Classify machine failures from sensor measurements.
Strengths	Fast; strong nonlinear baseline; handles interactions.
Limitations	Less interpretable; randomness may not help every dataset.
Practice project	Predictive Maintenance: compare Decision Tree, Random Forest, and Extra Trees using the same CV folds.
Useful metrics	F1/ROC-AUC or RMSE.
Interview cue	Difference between Random Forest and Extra Trees.

Gradient Boosting

Family	Supervised • Ensemble
Core idea	Builds trees sequentially, each correcting errors from the previous ensemble.
When to use	Structured/tabular data where predictive accuracy is important.
Why use it	Can model complex nonlinear relationships with strong predictive performance.
Simple example	Predict house prices or customer churn from tabular features.
Strengths	Powerful; handles nonlinear relationships; flexible loss functions.
Limitations	Can overfit; sequential training is slower than bagging; tuning matters.
Practice project	House Price Boosting: tune <code>learning_rate</code> , <code>n_estimators</code> , <code>max_depth</code> , and <code>subsample</code> ; compare against Random Forest.
Useful metrics	RMSE/MAE/R ² or F1/ROC-AUC.
Interview cue	Explain boosting as error correction and the learning-rate/tree-count tradeoff.

XGBoost

Family	Supervised • Gradient Boosting
Core idea	Optimized gradient-boosted trees with regularization and efficient implementation.
When to use	High-performance tabular classification/regression with mixed feature types and nonlinear relationships.
Why use it	Excellent practical accuracy, regularization, missing-value handling, and mature tooling.
Simple example	Predict loan default probability from customer and financial features.
Strengths	Strong tabular performance; regularization; scalable; handles nonlinearities.
Limitations	Many hyperparameters; can overfit; less transparent than linear models.
Practice project	Credit Default Prediction: build preprocessing + XGBoost, tune with randomized search, evaluate ROC-AUC/PR-AUC, and use SHAP for explanations.
Useful metrics	ROC-AUC, PR-AUC, F1; RMSE/MAE.
Interview cue	Know <code>learning_rate</code> , <code>n_estimators</code> , <code>max_depth</code> , <code>subsample</code> , <code>colsample_bytree</code> .

LightGBM

Family	Supervised • Gradient Boosting
Core idea	Gradient boosting optimized for speed and memory using histogram-based learning and leaf-wise growth.
When to use	Large tabular datasets where training speed and performance matter.
Why use it	Often trains quickly and handles large feature spaces efficiently.
Simple example	Predict e-commerce purchase probability from millions of sessions.

Strengths	Fast; memory-efficient; powerful on tabular data.
Limitations	Leaf-wise growth can overfit; categorical handling needs care depending on API/workflow.
Practice project	E-commerce Conversion Model: train LightGBM, tune num_leaves/min_child_samples, compare speed and AUC with XGBoost.
Useful metrics	ROC-AUC, PR-AUC, log loss; RMSE for regression.
Interview cue	Explain leaf-wise vs level-wise tree growth.

CatBoost

Family	Supervised • Gradient Boosting
Core idea	Gradient boosting designed with particularly strong handling of categorical features.
When to use	Datasets with many categorical columns and limited desire for extensive manual encoding.
Why use it	Reduces preprocessing work and can model category interactions effectively.
Simple example	Predict hotel booking cancellation using country, market segment, room type, lead time, and other fields.
Strengths	Strong categorical handling; good defaults; useful on mixed tabular data.
Limitations	Can be slower/heavier in some setups; still needs careful validation.
Practice project	Hotel Cancellation Prediction: compare native categorical handling with one-hot + XGBoost and inspect category importance.
Useful metrics	ROC-AUC, F1, log loss.
Interview cue	Explain ordered target statistics at a high level and why leakage prevention matters.

Support Vector Machine (SVM)

Family	Supervised • Classification/Regression
Core idea	Finds a margin-maximizing boundary; kernels can create nonlinear boundaries.
When to use	Small/medium datasets with high-dimensional features, especially when a clear margin exists.
Why use it	Effective in high-dimensional spaces and can use kernels for nonlinear patterns.
Simple example	Classify text documents or image features into categories.
Strengths	Strong in high dimensions; kernel trick; margin-based generalization.
Limitations	Scaling required; kernel SVM can be expensive on large datasets.
Practice project	Image/Document Classifier: standardize features, compare linear SVM vs RBF SVM, tune C and gamma.
Useful metrics	F1, accuracy, ROC-AUC.
Interview cue	Know C, margin, support vectors, kernels, and gamma.

K-Means Clustering

Family	Unsupervised • Clustering
Core idea	Partitions observations into k clusters by minimizing within-cluster squared distance.
When to use	You have unlabeled data and expect compact, roughly spherical groups.
Why use it	Simple, fast, and easy to communicate to business stakeholders.
Simple example	Segment customers into groups using spending, frequency, and recency.
Strengths	Fast; scalable; simple; easy to visualize.
Limitations	Must choose k; sensitive to scaling/outliers; assumes roughly spherical clusters.
Practice project	Customer Segmentation: RFM features → scaling → choose k with elbow/silhouette → profile each cluster → recommend marketing actions.
Useful metrics	Silhouette score, inertia, cluster profiles.
Interview cue	Why scaling matters and why k-means can fail on non-spherical clusters.

C. Unsupervised Learning & Dimensionality Reduction

Hierarchical Clustering

Family	Unsupervised • Clustering
Core idea	Builds a hierarchy of nested clusters represented by a dendrogram.
When to use	Small/medium datasets where you want to inspect cluster structure at multiple levels.
Why use it	Does not require committing to one k at the start and gives a visual hierarchy.
Simple example	Group products or customers into nested similarity segments.
Strengths	Dendrogram is informative; flexible linkage choices.
Limitations	Can be computationally expensive; sensitive to linkage/distance choices.
Practice project	Product Taxonomy Discovery: standardize product attributes, build dendrogram, select cut level, and compare with K-Means.
Useful metrics	Silhouette score; cophenetic correlation where appropriate.
Interview cue	Know agglomerative vs divisive and single/complete/average/Ward linkage.

DBSCAN

Family	Unsupervised • Clustering
Core idea	Groups dense regions and labels sparse points as noise.
When to use	Clusters are non-spherical and you expect outliers/noise.
Why use it	Finds arbitrary-shaped clusters and identifies noise without preselecting k.
Simple example	Detect geographic hotspots or unusual machine operating regimes.
Strengths	Arbitrary shapes; detects noise; no k required.
Limitations	Sensitive to eps/min_samples; struggles when cluster densities vary greatly.
Practice project	GPS Hotspot Detection: cluster delivery locations with DBSCAN, map clusters, and investigate noise points.
Useful metrics	Silhouette score with care; cluster counts; noise percentage.
Interview cue	Explain eps, min_samples, core/border/noise points.

Gaussian Mixture Model (GMM)

Family	Unsupervised • Probabilistic Clustering
Core idea	Represents data as a mixture of Gaussian distributions and gives soft cluster probabilities.
When to use	Clusters overlap and you want probabilistic membership rather than hard assignments.
Why use it	Soft assignments are useful when an observation can plausibly belong to multiple groups.
Simple example	Customer may be 70% segment A and 30% segment B.
Strengths	Probabilistic; flexible elliptical clusters; soft membership.

Limitations	Sensitive to initialization; Gaussian assumption; can overfit with too many components.
Practice project	Soft Customer Segmentation: fit GMMs with different component counts, select using BIC/AIC, and analyze membership probabilities.
Useful metrics	BIC/AIC, log-likelihood, silhouette as supplementary.
Interview cue	Difference between K-Means hard assignments and GMM soft assignments.

PCA

Family	Unsupervised • Dimensionality Reduction
Core idea	Rotates features into orthogonal directions that capture maximum variance.
When to use	Many correlated numeric features, visualization, noise reduction, or faster downstream modeling.
Why use it	Compresses information into fewer dimensions while preserving as much variance as possible.
Simple example	Reduce 100 sensor features to 10 components before classification.
Strengths	Useful compression; visualization; reduces multicollinearity.
Limitations	Components are less interpretable; variance is not always predictive information.
Practice project	Sensor Fault Visualization: standardize 50 sensor features, PCA to 2D, visualize classes, then compare classifier speed with/without PCA.
Useful metrics	Explained variance ratio; downstream model metric.
Interview cue	PCA is sensitive to scaling; fit PCA only on training data to avoid leakage.

LDA (Linear Discriminant Analysis)

Family	Supervised • Dimensionality Reduction/Classification
Core idea	Finds projections that separate known classes while reducing within-class variance.
When to use	You have labeled classes and want dimensionality reduction or a linear classifier.
Why use it	Uses class labels directly, unlike PCA.
Simple example	Project face or document features into a low-dimensional space that separates classes.
Strengths	Supervised; interpretable linear projection; useful for small/medium data.
Limitations	Assumptions may not hold; limited for complex nonlinear boundaries.
Practice project	Flower/Face Classification: compare PCA+Logistic Regression with LDA on the same dataset.
Useful metrics	Accuracy/F1; explained discriminative structure.
Interview cue	PCA maximizes variance; LDA maximizes class separation.

Isolation Forest

Family	Unsupervised • Anomaly Detection
Core idea	Randomly partitions data; anomalies are isolated in fewer splits.

When to use	Detecting unusual transactions, logs, sensor readings, or behavior without many labeled anomalies.
Why use it	Works well for high-dimensional tabular anomaly detection and does not require distance calculations like some methods.
Simple example	Flag unusually large or unusual banking transactions.
Strengths	Scalable; no labels required; useful for high-dimensional data.
Limitations	Contamination and feature representation matter; anomaly score needs interpretation.
Practice project	Fraud Screening: train Isolation Forest on mostly normal transactions, review top anomalies, and measure precision on a labeled holdout if available.
Useful metrics	Precision/recall on labeled anomalies; PR-AUC; top-k precision.
Interview cue	Know contamination and why anomaly detection is usually class-imbalanced.

D. Anomaly, Semi-Supervised & Ensemble Methods

Semi-Supervised Learning / Self-Training

Family	Semi-supervised
Core idea	Learns from a small labeled set plus a larger unlabeled set, often adding high-confidence pseudo-labels.
When to use	Labels are expensive but unlabeled examples are abundant.
Why use it	Can improve coverage without manually labeling every example.
Simple example	Classify support tickets when only 10% are manually labeled.
Strengths	Uses unlabeled data; can reduce labeling cost.
Limitations	Wrong pseudo-labels can reinforce errors; confidence thresholds matter.
Practice project	Support Ticket Classification: train a classifier on labeled tickets, pseudo-label only high-confidence cases, retrain, and compare to supervised-only baseline.
Useful metrics	F1, precision/recall on a strictly held-out labeled test set.
Interview cue	Never evaluate pseudo-label performance on data used to generate pseudo-labels.

AdaBoost

Family	Supervised • Ensemble
Core idea	Sequentially emphasizes examples that previous weak learners misclassified.
When to use	Small/medium tabular problems where simple weak learners can be combined effectively.
Why use it	Turns weak learners into a stronger ensemble by focusing on difficult observations.
Simple example	Classify whether a transaction is legitimate using shallow decision trees.
Strengths	Conceptually simple; strong on some clean tabular datasets.

Limitations	Sensitive to noisy/outlier-heavy data; tuning required.
Practice project	Transaction Classification: compare Decision Stump, AdaBoost, Random Forest, and Gradient Boosting under the same split.
Useful metrics	F1, ROC-AUC, PR-AUC.
Interview cue	Know weak learners, sample weights, and boosting vs bagging.

Voting Ensemble

Family	Supervised • Ensemble
Core idea	Combines predictions from several different models by hard votes or probability averaging.
When to use	Different models capture different errors and you want a simple ensemble.
Why use it	Combines complementary inductive biases without building a complex stacking system.
Simple example	Combine Logistic Regression, Random Forest, and SVM for customer churn.
Strengths	Simple; can improve robustness.
Limitations	More models mean more compute and maintenance; poorly chosen models add little value.
Practice project	Churn Voting Classifier: train three diverse models, compare hard vs soft voting, and analyze whether gains justify complexity.
Useful metrics	F1, ROC-AUC, calibration metrics.
Interview cue	Soft voting requires meaningful probability estimates.

E. Time Series & Recommendation

ARIMA

Family	Time Series • Forecasting
Core idea	Models autoregression, differencing, and moving-average error structure.
When to use	Univariate time series with temporal dependence and relatively stable patterns after differencing.
Why use it	Strong classical baseline for many forecasting problems.
Simple example	Forecast monthly electricity demand or sales.
Strengths	Interpretable; strong classical baseline; works with limited data.
Limitations	Requires stationarity-oriented diagnostics; struggles with complex nonlinear patterns and many exogenous variables unless extended.
Practice project	Monthly Sales Forecast: stationarity test, ACF/PACF, tune p,d,q, walk-forward validation, and compare with a naive baseline.
Useful metrics	MAE, RMSE, MAPE/sMAPE where appropriate.
Interview cue	Never randomly shuffle time series; use chronological validation.

Exponential Smoothing / Holt-Winters

Family	Time Series • Forecasting
Core idea	Uses weighted averages with more weight on recent observations; can model level, trend, and seasonality.
When to use	Forecasting data with level/trend/seasonal patterns and a relatively stable structure.
Why use it	Simple, fast, and strong for many business series.
Simple example	Forecast weekly retail demand with trend and seasonality.
Strengths	Fast; interpretable; excellent baseline.
Limitations	Less suitable for complex nonlinear patterns or many external predictors.
Practice project	Retail Demand Forecast: compare simple exponential smoothing, Holt, and Holt-Winters with rolling-origin validation.
Useful metrics	MAE, RMSE, sMAPE.
Interview cue	Know level, trend, seasonality, and additive vs multiplicative seasonality.

Collaborative Filtering

Family	Unsupervised/Recommendation
Core idea	Uses user-item interaction patterns to recommend items based on similar users or items.
When to use	You have interaction data such as ratings, clicks, views, purchases, or watch history.
Why use it	Personalizes recommendations without requiring detailed item content.
Simple example	Recommend movies based on what similar users watched/rated.
Strengths	Personalized; can exploit behavioral data.

Limitations	Cold start; sparse matrices; popularity bias.
Practice project	Movie Recommendation System: build user-item matrix, implement item-based similarity, evaluate Recall@K/Precision@K, and inspect cold-start behavior.
Useful metrics	Precision@K, Recall@K, MAP@K, NDCG@K.
Interview cue	Distinguish user-based vs item-based collaborative filtering.

Matrix Factorization / SVD for Recommenders

Family	Unsupervised/Recommendation
Core idea	Factorizes the user-item matrix into lower-dimensional user and item latent factors.
When to use	Large sparse recommendation matrices where latent preferences are useful.
Why use it	Captures hidden dimensions such as genre preference without manually defining them.
Simple example	Predict a user's missing movie rating from latent factors.
Strengths	Handles sparsity; compact latent representation; strong recommender baseline.
Limitations	Cold start; tuning and evaluation are important; classic SVD is not the same as truncated PCA.
Practice project	Movie Rating Recommender: train latent factors, predict held-out ratings, and compare against item-popularity and item-similarity baselines.
Useful metrics	RMSE for ratings; Precision@K/Recall@K for ranking.
Interview cue	Explain latent factors and why train/test must respect temporal/user leakage concerns.

F. Neural Network Foundation

Multilayer Perceptron (MLP)

Family	Supervised • Neural Network
Core idea	Stacks fully connected layers with nonlinear activations to learn complex mappings.
When to use	Nonlinear tabular or vector data when tree ensembles are not ideal or when you want a neural baseline.
Why use it	Learns nonlinear feature interactions through hidden layers.
Simple example	Predict customer churn from engineered numerical/categorical embeddings.
Strengths	Flexible; nonlinear; foundation for deeper networks.
Limitations	Needs scaling/tuning; can overfit; often not the first choice for ordinary tabular data.
Practice project	Churn Neural Network: build a small MLP with dropout, early stopping, and compare against XGBoost.
Useful metrics	F1, ROC-AUC; calibration; training/validation curves.

Interview cue	Know activation functions, backpropagation, epochs, batch size, learning rate, overfitting.
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G. One Project Per Algorithm — Practice Roadmap

For every project, follow the same professional structure: **EDA** → **data cleaning** → **baseline** → **preprocessing** → **train** → **cross-validation** → **tuning** → **error analysis** → **explanation** → **final report**.

Algorithm	Project deliverable	Minimum output
Linear / Ridge / Lasso	House or salary prediction	EDA + preprocessing + CV + coefficient analysis
Logistic Regression	Customer churn	ROC-AUC + threshold analysis + confusion matrix
KNN	Fruit/product similarity classifier	Scaling + k selection + decision boundary
Naive Bayes	Spam/sentiment classifier	TF-IDF + F1/PR-AUC + error analysis
Decision Tree	Loan approval	Tree visualization + pruning/tuning
Random Forest	Employee attrition	Feature importance + CV + comparison
Gradient Boosting / XGBoost	Credit default	Hyperparameter tuning + SHAP/error analysis
LightGBM / CatBoost	E-commerce conversion / hotel cancellation	Speed + performance comparison
SVM	Text/image classification	Scaling + kernel comparison
K-Means	Customer segmentation	Cluster profiles + business actions
Hierarchical	Product/customer hierarchy	Dendrogram + cluster selection
DBSCAN	GPS hotspot detection	Map clusters + analyze noise
GMM	Soft customer segmentation	BIC/AIC + membership probabilities
PCA	Sensor dimensionality reduction	Explained variance + 2D plot + downstream model
LDA	Class-separating projection	Compare against PCA + classifier
Isolation Forest	Fraud/anomaly screening	Top-k anomaly review + labeled evaluation if available
Semi-supervised	Support ticket labeling	Pseudo-label threshold + strict test set
AdaBoost / Voting	Transaction/churn ensemble	Compare single vs ensemble models
ARIMA / Holt-Winters	Sales/demand forecasting	Walk-forward validation + naive baseline

Collaborative Filtering / MF	Movie/product recommender	Top-K ranking metrics + cold-start analysis
MLP	Nonlinear churn classifier	Training curves + early stopping + baseline comparison

H. Interview Rapid Revision

- **Linear vs Logistic:** Linear predicts continuous values; Logistic predicts class probabilities.
- **Ridge vs Lasso:** Ridge uses L2 shrinkage; Lasso uses L1 and can set coefficients to zero.
- **Bagging vs Boosting:** Bagging trains models more independently and averages them; boosting trains sequentially to correct errors.
- **Random Forest vs XGBoost:** Random Forest is bagging; XGBoost is gradient boosting.
- **K-Means vs DBSCAN:** K-Means needs k and favors compact clusters; DBSCAN finds dense arbitrary shapes and noise.
- **PCA vs LDA:** PCA ignores labels and maximizes variance; LDA uses labels to maximize class separation.
- **Precision vs Recall:** Precision asks “of predicted positives, how many are correct?” Recall asks “of actual positives, how many did we find?”
- **Overfitting controls:** regularization, pruning, shallower trees, fewer features, more data, dropout/early stopping for neural networks, and proper cross-validation.

I. Suggested Portfolio Sequence

If your goal is Data Science / ML interviews, build these projects in increasing difficulty:

1. **House Price Prediction** — Linear/Ridge/Lasso + feature engineering.
2. **Customer Churn** — Logistic Regression + Random Forest + XGBoost.
3. **Spam/Sentiment** — TF-IDF + Naive Bayes + Logistic Regression/SVM.
4. **Customer Segmentation** — K-Means + DBSCAN + business profiling.
5. **Fraud/Anomaly Detection** — Isolation Forest + imbalance-aware evaluation.
6. **Sales Forecasting** — naive baseline + Holt-Winters + ARIMA.
7. **Recommendation System** — collaborative filtering + matrix factorization.
8. **End-to-End ML Project** — preprocessing pipeline, model registry/versioning, API, Docker, monitoring, and retraining plan.

Golden rule: A good ML project is not “I trained XGBoost.” It is a complete chain from business problem → data quality → baseline → model → correct validation → error analysis → explainability → deployment/monitoring.

End of guide • Keep this PDF as a revision sheet and use the project column as your implementation checklist.